ANALYSTLAB AFRICA

Week 4 — FinTrust Digital Bank: Test, Refine & Present

Field	Detail
Intern Name	Tatenda Farai Mujati
Track	Data Analytics
Week	Week 4 (final)

Deliverables map

Required deliverable	Where to find it
Final Power BI dashboard	FinTrust_Week4_Analytics_Dashboard.pbix (+ PDF export); design and review in Section 6
Final SQL scripts	FinTrust_Week4_Final_SQL_Analysis.sql (9 queries, executed and verified); summary in Section 5
Final Python notebook	FinTrust_Week4_Final_Analysis.ipynb (reproducible; 7 charts; automated checks and statistical tests)
KPI documentation	Section 4
Final business insights	Section 7
Recommendations	Section 9
Validation evidence	Sections 8 and 10 (plus notebook Section 4)
Data quality documentation	Section 3 (builds on the Week 2 Data Quality Report and Cleaning Log)
Final project documentation	This document

1. Executive Summary

FinTrust wants to turn its customer and transaction records into business intelligence that supports decisions about risk review, reliability and customer strategy. Over four weeks the Data Analytics track moved from understanding the problem (Week 1), through cleaning and first analysis (Week 2) and deeper SQL/Python/dashboard work (Week 3), to this final stage, where the emphasis is on testing whether the conclusions are actually true.

Every finding reported in Weeks 2–3 was re-tested with an appropriate statistical method and every headline figure was recomputed from the source data. The result is a smaller but more reliable set of conclusions:

- Three findings are strongly supported: risk-review is concentrated in Transfers (28.5%) and Cash Withdrawals (25.3%); international transactions are flagged about twice as often as domestic ones (36.9% vs 18.9%) in every transaction type; and transaction value is extremely concentrated — the top 10% of transactions carry 56% of all value.
- Five earlier statements did not survive testing and are corrected in this report: the "February dip" (a calendar artefact), the "narrowing" international gap (not statistically significant), USSD's "higher" average value (no real difference between channels), Web as the worst channel for failures (not distinguishable from the others), and above-average risk in the Premium/Student segments (no difference between segments).
- Three percentage figures in earlier narrative text were arithmetic errors (Lagos 36.1% of value is actually 33.9%; Mobile App 45.7% is 42.8%; Everyday 49.6% is 46.6%) and are corrected in Section 2.

2. Part A — Final Data & Analysis Review

2.1 Review checklist

Requirement	Evidence	Status
Data-quality issues addressed	13 of 13 automated checks pass (unique keys, no duplicates, no orphans, valid ranges, clean categories). The only gaps — 96 blank Device_Type and 96 blank Location cells (0.8% each) — are handled as 'Unknown' per the Week 2 cleaning log.	Confirmed
Calculations are accurate	All six KPIs reconcile exactly across pandas, SQL and the dashboard cards. A narrative audit found three percentage errors in earlier write-ups (table 2.2, items 1–3), now corrected.	Corrected
SQL queries produce expected results	All 9 final queries were executed end to end in SQLite; FQ1 matches the pandas KPIs; every interpretation quotes the executed output.	Confirmed
Python analysis is reproducible	The notebook runs top to bottom from the two data files with relative paths and no hidden state; its outputs were produced by executing the code shown.	Confirmed

Requirement	Evidence	Status
KPIs are clearly defined	Section 4 documents definition, DAX/SQL formula, final value and validation for each KPI.	Confirmed
Findings are supported by evidence	10 findings re-tested: 5 supported, 5 revised (Section 8).	Confirmed

2.2 Corrections and improvements log

Everything below was reported in an earlier week and has been corrected or withdrawn in this final version.

#	Reported earlier	What testing / re-calculation showed	Correction
1	Lagos = 36.1% of transaction value (Wk2 SQL, Wk2 doc, Wk3 doc)	NGN 189.9m of 560.5m = 33.9%. The 36.1% was an arithmetic slip; the volume share (32.6%) was right.	Corrected to 33.9%
2	Mobile App = 45.7% of value (Wk2 SQL)	NGN 240.1m of 560.5m = 42.8% (volume share 42.5% was right).	Corrected to 42.8%
3	Everyday segment = 49.6% of value (Wk2 SQL)	NGN 261.5m of 560.5m = 46.6%.	Corrected to 46.6%
4	February dip in value (Wk3 SQL AQ2, notebook, doc). The Wk3 notebook also claimed the dip was larger than the shorter month explains.	Per active day, volume is identical (133.3 vs 133.4) and value is +3.3% in February. The weekly chart also included two partial weeks (first: 4 days; last: 2 days).	REVISED — no dip
5	International risk gap is narrowing, 41.6% → 33.2% (Wk3)	Jan vs Mar difference: z = 1.61, p = 0.107; monthly 95% intervals overlap (±7 pts, ~160 transactions a month).	REVISED — gap is stable within noise
6	USSD has the highest average transaction value (Wk2)	Channel means are NGN 43.6k–48.1k; ANOVA p = 0.39, Kruskal-Wallis p = 0.75. Medians are ~NGN 9.5k–10.7k.	REVISED — no real difference
7	Web has the highest value-at-risk, so prioritise Web (Wk3 AQ6, Finding 2, Rec 2)	Failure rate by channel: chi-square p = 0.07; bootstrap intervals overlap; ATM vs rest p = 0.074.	Withdrawn — failures are platform-wide
8	Premium and Student have above-average risk (Wk3 AQ3)	Segment risk rates 19.3%–20.3%; chi-square p = 0.75.	REVISED — no difference

#	Reported earlier	What testing / re-calculation showed	Correction
9	Top customers are not 'Premium' / high-value customers sit in every segment (Wk2 Finding 5, Wk3 Finding 5, Rec 5)	Segments do not differ on customer value (Kruskal p = 0.46), activity (7.9–8.2 transactions) or engagement (p = 0.54). 3 of the top 5 being Everyday is expected as Everyday is 47% of customers.	REVISED — segment labels carry no value signal
10	Wk3 recommendations 1–5	Rec 1 (use Location, not City, for transaction geography) retained. Rec 2 withdrawn. Rec 3 retained without the 'narrowing' claim. Rec 4 (value-weighted monitoring) retained and strengthened. Rec 5 reframed as 'rebuild segmentation'.	Updated (Section 9)

3. Data Quality Documentation

Full detail is in the Week 2 Data Quality Report (PDF) and Data Cleaning Log (Excel). This section is the final status, re-verified in Week 4 with automated checks that can be re-run in the notebook.

Issue	Dataset / field	Records	Handling	Final status
Missing values	Transaction: Device_Type	96 of 12,000 (0.8%)	Filled with 'Unknown' (log CL-01)	Closed
Missing values	Transaction: Location	96 of 12,000 (0.8%); 4 rows missing both	Filled with 'Unknown' (log CL-02)	Closed
Naming collision	Account_Type 'Premium' vs Customer_Segment 'Premium'	229 customers hold a Premium account type	No data changed; always qualify the field name in queries and filters	Accepted (documented)
Week 1 hypothesis tested	Channel labels	12,000	Hypothesised inconsistent labels; full file has exactly 5 clean values	Closed (hypothesis rejected)
Skewed distribution	Amount_NGN	~12% above the blanket IQR fence	Legitimate variation across types (NGN 100 – 693,454); not removed	Accepted (documented)
Key integrity	Customer_ID / Transaction_ID	0 duplicates, 0 orphans	None needed	Confirmed clean

New observation this week: the synthetic data is unusually uniform

Daily volume is 133.3 ± 0.5 transactions on every one of the 90 days, hourly volume is exactly 500 transactions in each of the 24 hours, international transactions are ~4% of every transaction type, and transaction location is statistically unrelated to the customer's home city. Real banking data does not look like this. It is a feature of how the dataset was generated and means that time-of-day, day-of-week and seasonal analysis cannot reveal real behaviour here. It is recorded as a limitation, not a defect to fix.

4. KPI Documentation

Each KPI was calculated independently in pandas and SQL and compared with the dashboard card; all agree exactly.

KPI	Definition	Formula (DAX / SQL logic)	Final value
Total Customers	Distinct customers in the base	DISTINCTCOUNT(Customer_Data[Customer_ID])	1,500
Total Transactions	All recorded transactions	COUNTROWS(Transaction_Data)	12,000
Total Transaction Value	Sum of transaction amounts	SUM(Transaction_Data[Amount_NGN])	₦560,477,354.85
Average Transaction Value	Mean amount (right-skewed; read with the median)	AVERAGE(Transaction_Data[Amount_NGN])	₦46,706.45
Median Transaction Value	Typical transaction (not pulled up by very large ones)	MEDIAN(Transaction_Data[Amount_NGN])	₦10,305.94
Transaction Success Rate	Successful ÷ all transactions	DIVIDE(CALCULATE(COUNTROWS(Transaction_Data), Transaction_Data[Transaction_Status]="Successful"), COUNTROWS(Transaction_Data))	90.47%
Failed + Reversed Rate	Share of transactions that failed or reversed	DIVIDE(CALCULATE(COUNTROWS(Transaction_Data), Transaction_Data[Transaction_Status] IN {"Failed","Reversed"}), COUNTROWS(Transaction_Data))	7.97% (956)

KPI	Definition	Formula (DAX / SQL logic)	Final value
Value at Risk %	Value of failed/reversed transactions ÷ total value	DIVIDE(CALCULATE(SUM(Transaction_Data[Amount_NGN]), Transaction_Data[Transaction_Status] IN {"Failed","Reversed"}), SUM(Transaction_Data[Amount_NGN]))	7.22% (₦40.45m)
Risk Review Rate	Flagged ÷ all transactions (synthetic label)	DIVIDE(CALCULATE(COUNTROWS(Transaction_Data), Transaction_Data[Risk_Review_Flag]="Yes"), COUNTROWS(Transaction_Data))	19.60%
Average Daily Value (new)	Total value per active day — lets months of different length be compared fairly	Add column: Txn Date = DATE(YEAR([Transaction_DateTime]), MONTH([Transaction_DateTime]), DAY([Transaction_DateTime])); measure: DIVIDE(SUM(Transaction_Data[Amount_NGN]), DISTINCTCOUNT(Transaction_Data[Txn Date]))	₦6,227,526 (90 days)

5. Part B — Final SQL Analysis

The nine final queries are in FinTrust_Week4_Final_SQL_Analysis.sql. They were selected from the 24 queries written across Weeks 2–4 as the most decision-relevant and each one was executed with its results quoted in the file.

ID	Techniques	What it reveals	Versus earlier weeks
FQ1	Aggregation, CASE, subquery	The six headline KPIs in one statement	Reconciled with pandas/Power BI
FQ2	CTE, RANK()	Transfer (28.49%) and Cash Withdrawal (25.31%) are ranked highest for risk-review; the other four types sit below the 19.6% average	Validated
FQ3	Conditional aggregation	International is higher within every type (+14 to +23 pts) at a constant ~4% share, so it is not a type-mix artefact	Validated; ‘narrowing’ withdrawn
FQ4	CTE, NTILE(), cumulative window	Top decile = 56.15% of value; bottom half of transactions = 3.4%	Validated (CI 55.2–57.0%)
FQ5	NTILE()	Risk-review is flat (17.5–18.4%) for four quintiles, then 26.75% for transactions of ~₦68k and above	New

ID	Techniques	What it reveals	Versus earlier weeks
FQ6	CTE, LAG(), COUNT DISTINCT	Per active day, volume is identical across months and value is +3.3% in Feb	Corrects Wk3 ‘Feb dip’
FQ7	Conditional aggregation	Failure/reversal rate 6.9–8.7% by channel; ranking flips between count and value; differences not significant	Replaces ‘Web is worst’
FQ8	LEFT JOIN, CTE, window AVG	Segments differ by only ~6% in value per customer and by <0.4 in transactions per customer	Replaces two segment claims
FQ9	Three CTEs, JOINs	Lagos: 32.5% of customers, 33.9% of value by home city, 11.5% of value by transaction location	Confirms and sharpens Wk3 correction

6. Part C — Final Power BI Dashboard

The final dashboard builds on the Week 2 and Week 3 versions rather than starting again. Its structure follows the business story: how big is the business, how is it trending, where is the risk, how reliable is it and who and where are the customers.

6.1 Final design

Section	Visuals	Question answered
KPI row	Total Customers · Total Transactions · Total Value · Average Value · Success Rate · Risk Review Rate · Average Daily Value	How large and healthy is the business?
Trend	Value per active day by month (column chart)	Is activity changing, measured fairly across months of different length?
Transactions & channels	Transaction type by channel (stacked column); channel share	How is activity split by product and channel?
Risk review	Risk Review Rate by Transaction Type; Risk Review Rate by Channel	Where is the risk-review label concentrated? (strongest signal: transaction type)
Transaction status	Transactions by status; Failed + Reversed Rate card	How reliable is the platform?
Customers & geography	Average engagement by segment; Value by Home City (customer base); Value by Transaction Location	Who are the customers, and where do they live versus transact?

Section	Visuals	Question answered
Slicers	Date range · Customer Segment · Channel · Account Status · Transaction Type (each with Select All)	Slice any view; Power BI cross-filtering links all visuals

6.2 Final review against the Week 4 criteria

Criterion	What was checked	Outcome / refinement
KPI accuracy	Each card compared with the independent pandas and SQL values	All match (Section 4). New Average Daily Value card added so months compare fairly.
Visual clarity	One question per visual; no duplicated charts	The noisy daily line and the weekly chart (which included two partial weeks) were replaced by one value-per-day-by-month chart.
Titles and labels	Plain-language titles	Geography charts explicitly labelled ‘Home City’ vs ‘Transaction Location’.
Filters / slicers	Each slicer tested alone and in combination; Select All enabled	Cross-filtering confirmed (Week 3 test: selecting a channel updates every card and visual).
Consistency	Same colours for the same categories; sections aligned in a grid	Teal for neutral measures, coral reserved for risk.
User experience	Reads top to bottom as a story; section headers added	Risk Review Rate by Transaction Type added because it is the strongest finding.

7. Part D — Final Business Insights

Finding	Evidence	Business meaning	Recommended action
1. Risk-review is concentrated in Transfers and Cash Withdrawals	28.5% and 25.3% vs 19.6% overall; the other four types are 12.8%–16.2%. Chi-square $p < 1e-70$; Cramér’s $V = 0.17$ (the strongest association found).	Review effort spread evenly across types is largely spent on lower-rate activity. Transaction type is the best single guide to where flags occur.	Direct review capacity first at Transfers and Cash Withdrawals.

Finding	Evidence	Business meaning	Recommended action
2. International transactions are flagged ~2x as often, regardless of type	36.9% vs 18.9% ($p < 1e-21$). Higher in all six types (+14 to +23 pts); ~4% of every type. Month-to-month change is within noise ($p = 0.107$).	A real, stable, independent risk indicator on a small slice (480 transactions, 4% of volume).	Keep 'international' as a review criterion on top of type; do not assume the gap is closing.
3. Value is extremely concentrated, and the largest transactions are flagged most	Top 10% of transactions = 56.15% of value (CI 55.2–57.0%; Gini 0.735); bottom 50% = 3.4%. Flag rate 26.8% in the top 20% by amount (≥ ~₦68k) vs 17.8% below.	Equal-weighted monitoring covers little money per unit of effort. The costliest errors sit in the top fifth of transactions.	Introduce value-tiered monitoring: closest scrutiny above ~₦68k.
4. Failures and reversals are platform-wide, not a channel problem	956 transactions (7.97%), ₦40.45m (7.22% of value). By channel 6.9–8.7% ($p = 0.07$), segment ($p = 0.08$), account status ($p = 0.89$), city ($p = 0.98$): no clear hotspot.	Fixing one 'worst' channel would be guesswork; the evidence points to a common cause or to random variation.	Track failure rate weekly as a KPI; capture a failure-reason code to find causes.
5. Customer segments do not differ in value, activity, engagement or risk	Value per customer within ~6% ($p = 0.46$); 7.9–8.2 transactions each (similar); engagement 66.7–68.4 ($p = 0.54$); risk 19.3–20.3% ($p = 0.75$).	The segment labels are not separating customers by behaviour, so campaigns aimed at 'Premium' or 'Student' are not targeting different customers.	Rebuild segmentation on behaviour: value tier, frequency tier, transaction mix.
6. There is no February dip; months must be compared per day	Per active day: 133.3 / 133.4 / 133.3 transactions; value ₦6.08m / ₦6.28m / ₦6.33m.	A raw monthly total always makes a 28-day month look weaker; acting on it would be a false alarm.	Report period comparisons per active day (Average Daily Value).
7. Lagos is a customer-base fact, not a transaction-load fact	Lagos = 32.5% of customers and 33.9% of value by home city (proportional); by transaction location every city is 11–15% ($p = 0.34$); location is unrelated to home city ($p = 0.67$).	Home city answers 'where are our customers?'; Location answers 'where is activity recorded?'. They must not be interchanged.	Use City for acquisition/marketing; confirm with the data owner what Location represents before using it for infrastructure planning.

8. Part E — Final Validation

Ten findings were re-tested. Results can be reproduced in Section 4 of the notebook. “p” is the probability of seeing a difference this large if there were really none; values above 0.05 are treated as not distinguishable from chance.

Finding	Validation method	Result	Conclusion
Risk-review is highest for Transfers and Cash Withdrawals	Chi-square test; Cramér’s V; 95% Wilson intervals	chi2 = 358, p = 2.9e-75, V = 0.173; intervals of the top two do not overlap the other four	SUPPORTED
International transactions are flagged ~2x as often	Chi-square; stratified by transaction type	36.9% vs 18.9%, p = 4e-22; higher in all six types	SUPPORTED
The international gap is narrowing	Two-proportion z-test, Jan vs Mar; monthly intervals	z = 1.61, p = 0.107; ±7 pts intervals overlap	REVISED — not significant; gap is stable within noise
Top 10% of transactions carry ~56% of value	Bootstrap (1,000 resamples); Gini; composition of top decile	56.15% (CI 55.2–57.0%); Gini 0.735; top decile = 51% Transfers, 26% Deposits	SUPPORTED
February shows a dip in value	Normalise by active days; check partial weeks	Raw −6.7% but per-day +3.3%; txns/day 133.3 vs 133.4	REVISED — calendar artefact, no dip
USSD has the highest average transaction value	ANOVA and Kruskal-Wallis; medians	p = 0.39 and 0.75; channel means within ~10%	REVISED — no real difference between channels
Web has the highest share of value lost to failures	Chi-square on failure rate; bootstrap intervals; ATM vs rest	p = 0.070; intervals overlap; ATM vs rest p = 0.074	REVISED — no clear worst channel
Premium and Student segments carry above-average risk	Chi-square across four segments	Rates 19.3%–20.3%, p = 0.75	REVISED — no difference
Segment labels do not identify high-value customers	Kruskal-Wallis on customer total value; top-5% composition	p = 0.46; top 5% = 41% Everyday vs 47% of base	SUPPORTED (reframed: segments carry no value signal)

Finding	Validation method	Result	Conclusion
Home city and transaction location measure different things	Chi-square independence; match rate vs chance; uniformity test	p = 0.67; fields agree 12.7% vs 12.4% expected by chance; location uniform p = 0.34	SUPPORTED

9. Recommendations

Priority	Recommendation	Basis
High	Concentrate risk-review capacity on Transfers and Cash Withdrawals, and apply 'international' as an additional criterion within every type.	Insights 1–2
High	Adopt value-tiered monitoring: treat transactions above roughly ₦68,000 (top 20%) as the highest-scrutiny tier — they carry most of the money and the highest flag rate.	Insight 3
Medium	Treat failures and reversals (7.97% of transactions, ₦40.5m) as a platform-wide KPI tracked weekly, and add a failure-reason field so root causes can be identified. Do not single out one channel on current evidence.	Insight 4
Medium	Rebuild customer segmentation on behaviour (value tier, frequency tier, product mix) because the current labels do not differentiate customers.	Insight 5
Medium	Standardise geography definitions: use home City for customer-base decisions and Location for transaction activity, after confirming with the data owner what Location records.	Insight 7
Process	Report period comparisons per active day, and show a sample size and confidence interval before announcing any difference between groups.	Insights 2, 4, 6

10. Final Testing & Validation Evidence

Monthly trend

Stage	Detail
Final Output	Week 3 dashboard weekly-trend visual and SQL AQ2 reported a February dip of −6.7% in value.
Test	Divided each month by its number of active days; inspected the first and last weeks of the weekly series.
Finding	Transactions per day are identical (133.3 / 133.4 / 133.3) and value per day is +3.3% in February. The weekly series contained two partial weeks (4 days and 2 days) that exaggerated its low points.

Stage	Detail
Refinement	Replaced the trend visual with Value per Active Day by month; added the Average Daily Value KPI; rewrote the insight (Section 7, #6).
Re-test	Recomputed per-day figures independently in SQL (FQ6) and pandas (notebook Chart 5) — identical results.
Final Result	A trend view that cannot be distorted by month length; a false alarm removed before reaching a recommendation.

Channel and segment claims

Stage	Detail
Final Output	Earlier findings: USSD highest average value; Web highest value-at-risk; Premium/Student above-average risk; high-value customers hidden in non-Premium segments.
Test	ANOVA and Kruskal-Wallis (channel averages); chi-square and bootstrap intervals (failure rate by channel); chi-square (segment risk); Kruskal-Wallis (segment value).
Finding	None of the differences is statistically distinguishable from random variation (p = 0.39/0.75, 0.07, 0.75, 0.46).
Refinement	Withdrew the four claims; replaced them with two evidence-backed insights — failures are platform-wide (#4) and segments do not differ (#5) — and updated the recommendations.
Re-test	Each conclusion was confirmed with a second method (parametric and non-parametric; chi-square and bootstrap).
Final Result	Recommendations now point at what the data supports (transaction type, international status, value tiers) instead of differences that are noise.

Narrative figures

Stage	Detail
Final Output	Percentages quoted in the Week 2–3 text (Lagos 36.1% of value, Mobile App 45.7%, Everyday 49.6%).
Test	Recalculated every value share from the source data and compared with the text.
Finding	Three figures were wrong: the correct shares are 33.9%, 42.8% and 46.6%. The table figures themselves had been correct — the error was in the percentage written beside them.
Refinement	Corrected the figures (Section 2.2) and generated the shares by query (FQ9) rather than typing them by hand.

Stage	Detail
Re-test	SQL FQ9 and pandas give the same values.
Final Result	Every number in the final documents is traceable to an executed query.

11. Final Integration

This is a Data Analytics submission; the table states honestly what it contributes and what it does not claim.

Stage in the FinTrust flow	Data Analytics contribution	Integration status
Customer & transaction data	Cleaned datasets, data-quality log and report, documented limitations	Delivered
Analytics findings & KPIs	10 KPI definitions, 7 validated insights, 6 recommendations, the dashboard and the SQL/Python evidence	Delivered — this submission
Predictive risk intelligence (Data Science)	Evidence on which fields relate to the synthetic Risk_Review_Flag (below). No predictive model was built in this track.	No output received from or used from this track no integration is claimed
ML workflow / service (ML Engineering)	None directly; cleaned data and KPI definitions are available as inputs	No output received or used — no integration is claimed
Knowledge-based GenAI support	None directly; failure prevalence (7.97%) and status mix could inform which support scenarios matter most	No output received or used — no integration is claimed

Analytics evidence available to a downstream predictive model

These associations come from descriptive tests, not from a trained model and are offered as hypotheses for the Data Science track to test, not as conclusions about predictive power.

Field	Association with Risk_Review_Flag	Strength
Transaction_Type	Chi-square p < 1e-70; Transfer 28.5% ↔ Bill Payment 12.8%	Moderate (V = 0.17)
International_Transaction	36.9% vs 18.9%; holds within every type	Small–moderate (V = 0.09)
Amount_NGN (top quintile)	26.8% flagged above ~₦68k vs 17.8% below; flat below that	Threshold effect
Transaction_Status	Failed transactions flagged 27.0% vs 19.2% for Successful	Small (V = 0.04)

Field	Association with Risk_Review_Flag	Strength
Channel, Account_Type	p = 0.019 and 0.042; very small effects (V ≈ 0.03 and 0.02) — treat cautiously	Weak
Segment, Device_Type, Location, City, Account_Status, Gender, Income band, Preferred channel	No detectable association (all p > 0.2)	None

12. Assumptions, Limitations & Responsible Use

- FinTrust is fictional and all data is synthetic. This is a learning project, not a production banking system and no finding here should be used for real financial decisions.
- Risk_Review_Flag is a synthetic educational label. It is not a fraud determination and 'risk' in this report always means 'flagged for review' in that sense.
- The data covers one quarter (1 Jan – 31 Mar 2026) and is unusually uniform over time, so no seasonal, weekly or time-of-day conclusions can be drawn.
- The results are associations, not causes. A higher flag rate for Transfers does not show that Transfers cause review.
- Many statistical tests were run. At a 0.05 threshold roughly one in twenty could appear 'significant' by chance, so borderline results (Channel p = 0.019, Account_Type p = 0.042) are described as weak rather than relied on.
- There is no failure-reason field, so the analysis can show how often transactions fail but not why.
- The meaning of the Location field is not documented; it is statistically unrelated to home city, so conclusions about it are limited to what the data shows.
- 96 blank Device_Type and 96 blank Location cells were filled with 'Unknown' and are excluded from Location-based shares.
- Power BI results depend on the data model: if the Customer↔Transaction relationship is changed, KPIs will no longer match the validated values.
- No unsupported assumptions were carried into the final conclusions; where an earlier assumption failed testing it is listed in Section 2.2.

13. Lessons Learned

What worked well

1. Re-testing every finding before finalising caught five claims that were not real and three arithmetic slips.
2. Cross-checking with several tools (Excel, SQL, Python, Power BI) meant a number was only trusted when at least two agreed.
3. Keeping a corrections log made it possible to change my mind transparently instead of quietly.

Challenges

1. Power BI configuration

2. Reworking conclusions late in the project when earlier findings did not survive testing.

3. Synthetic data that is almost perfectly uniform over time, which limited the trend analysis that was originally planned.

Lessons

1. A pattern in a chart is a hypothesis, not a finding — check sample size and random variation before writing the sentence.

2. Normalise before comparing: per-day instead of per-month, rates instead of counts, medians alongside means.

3. Never type a computed number by hand into a report; generate it from the query that produced it.

Key improvements made in Week 4

- Added the Average Daily Value KPI and replaced the misleading trend views; added a Risk Review Rate by Transaction Type chart; separated Home City from Transaction Location.
- Replaced five unsupported findings with evidence-backed ones and updated the recommendations.
- Made the notebook fully reproducible with automated data-quality checks and a three-way KPI reconciliation.

Recommendations for future project work

- Agree metric definitions and field meanings (especially Location and Segment) with the data owner in week one.
- Build significance and confidence-interval checks into the first analysis rather than the last.
- Capture failure reasons and a longer date range in the source data so reliability and seasonality can be studied properly.